



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)

EXAMINATIONS - JAN/FEB 2023

**SUBJECT : ECE 2153 - DIGITAL SYSTEM DESIGN
(MAKEUP)**

DIGITAL SYSTEM DESIGN [ECE 2153]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Implement the following logic function using (i) 8:1 multiplexer (ii) 2:1 multiplexer.
 $F(A, B, C, D) = \sum m(1, 3, 4, 11, 12, 13, 14, 15)$ (4)
- 1B) Design a combinational circuit to convert the decimal numbers 0, 1, 4, 6, 7 and 8 in a 4-bit Excess 3 code input. (3)
- 1C) Implement the full adder using active low decoder. (3)
- 2A) Design a code converter that converts a 8421 BCD code into excess 3 code. The converter has four inputs A, B, C, D and four outputs f1, f2, f3 and f4. Implement the circuit using 3:8 decoder with active low outputs. (4)
- 2B) Derive the characteristic equation of S-R flipflop. Convert S-R flipflop to J-K flipflop. (3)
- 2C) Design a mod-6 Asynchronous up-counter using T Flipflops. (3)
- 3A) Illustrate the working of 4-bit universal shift register with the help of neat diagram register as per the Table Q3A

Table Q3A		
Control Input		Register Operation
S ₁	S ₀	
0	0	No change
0	1	Shift right
1	0	Shift left
1	1	Parallel load

- 3B) Construct a 2-bit synchronous down counter using T-flip flop. (3)
- 3C) Write a structural Verilog code for 8- to-1 multiplexer using 2-to-1 multiplexer. (3)
- 4A) Write a sequential Verilog code for 8 to 3 priority encoder. (4)
- 4B) Write structural Verilog code for 4-bit binary-to-gray code converter using XOR gate as basic building block. (3)
- 4C) Write a Verilog code for 3-input XOR gate using 2-input XOR gate as user defined primitive (UDP). (3)

- 5A) Implement a 4-bit SISO register using Xilinx CLB. (4)
- 5B) Implement $F = \sum m(1,3,5,6)$ using ACT-1 module. (3)
- 5C) Implement a BCD to XS-3 code conversion circuit using PROM. (3)

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